Points to Remember

- Binomial distribution is a discrete distribution.Its p.m.f is given by

$$P(X=x)=\begin{cases} nC_x p^x q^{n-x} & for\ x=0,1,2,...n \\ 0 & otherwise \end{cases}$$ where q=1-p

n,p are the parameters of the Binomial distribution.Then

(i)Mean = np; (ii)Variance = npq; (iii) Standard deviation = $\sqrt{npq}$

(iv)Skewness = $\frac{q-p}{\sqrt{npq}}$; (v)Kurtosis = $\frac{1-6pq}{npq}$

- Poisson distribution is a discrete distribution.Its p.m.f is given by

$P(X=x)=\frac{e^{-\lambda}\lambda^x}{x!}$, x= 0, 1, 2, ...∞,

λ is the parameter of the Poisson distribution.Then

(i) Mean = λ = variance (ii) Standard deviation = $\sqrt{\lambda}$

(iii) Skewness = $\frac{1}{\sqrt{\lambda}}$ (iv) Kurtosis = $\frac{1}{\lambda}$

- Rectangular or Uniform distribution is a continuous distribution.The p.d.f isgiven by

$$f(x)=\begin{cases} \frac{1}{b-a} & a<x<b \\ 0 & otherwise \end{cases}$$ a,b are the parameters of the rectangular distribution.Then

(i) Mean $\mu=\frac{a+b}{2}$ (ii) Variance $\sigma^2=\frac{(b-a)^2}{12}$

(iii) Median = $\frac{a+b}{2}$ (iv) Skewness = 0 (v)Kurtosis = $-\frac{6}{5}$

- A continuous random variable X with mean μ, and variance σ^2 as the parameters follows the Normal distribution. Its p.d.f is given by

$$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left\{\frac{x-\mu}{\sigma}\right\}^2}$$ where $-\infty<x<\infty$, $-\infty<\mu<\infty$ and $\sigma>0$

- In Normal distribution;

(i) Mean = Median = Mode = μ (ii) Modal height = $\frac{1}{\sigma\sqrt{2\pi}}$

(iii) Skewness β_1 = 0 (iv) Kurtosis β_2 = 3

- Area Properties of Normal Distribution

(i) P (- ∞<X<∞) = 1

(ii) P(μ - σ< X <μ + σ) = 0.6826

(iii) P (μ - 2σ< X < μ + 2σ) = 0.9544

(iv) P (μ - 3σ< X <μ + 3σ) = 0.9973